



黑客松 Work Shop

Vincent Chen

Mail address: vincent-ft.chen@advantest.com

Agenda

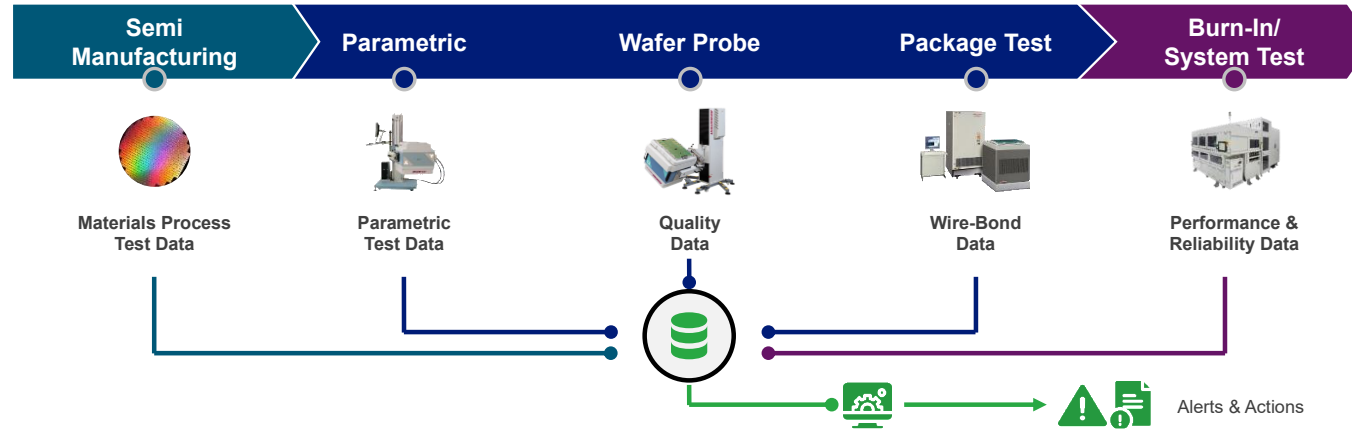
- **What is RTDI**
- **Gemini overview**
 - How to use Gemini
- **NEXUS ONEAPI overview**
- **Development environment**
- **Deployment and production run simulation**
 - Docker image deployment
 - Simulate production run
- **Dataset overview**

ACS Real-Time Data Infrastructure is all about...

Cross-Test-Floors Data Streaming



Time-to-market
ML Apps
Development



Automated ML
Models Deployment
to Production/OSATs

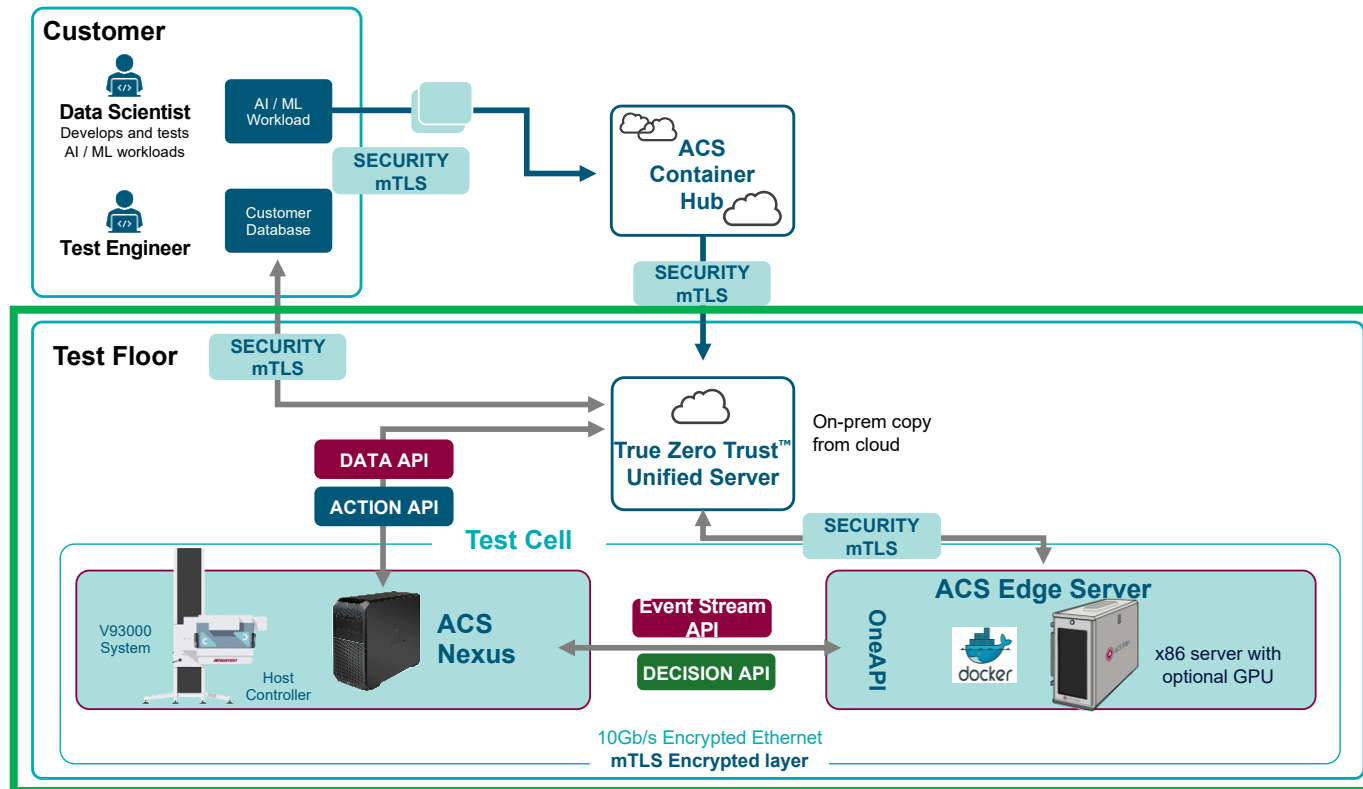
Data-Driven Decision-Making During Test



ACS RTDI™ Optimizes Time-to-ML Deployment



- ML Ops development shortens lead time
 - From 6M to 1M for new device
 - From 4Ws to 2Ws for replicating developed ML on new device
 - From 4 weeks to 2~3 days for new ML on same device



• Modular Test Program Development

- **Decouples** *Test Program* from *Data* and *ML Model* to minimize test program changes
- **Encapsulates** ML building blocks in APP to be **reused** by multiple test programs
- Enables **offloading** compute-intensive test program logic to run in parallel

• Advanced Debugging Capabilities

- **ACS Gemini™:** cloud-based simulation and debugging environment to accelerate ML development and deployment.

• Automated ML Models Deployment to OSATs

- **Independent ML release cycles**, faster updates and rollouts without disrupting production testing
- Enabling **rapid deployment and scaling of AI/ML models** across foundry and OSAT production floors
- **Real-time monitoring***: full control over application performance, system up-time, events, and logs

Agenda

- **What is RTDI**
- **Gemini overview**
 - How to use Gemini
- **NEXUS ONEAPI overview**
- **Development environment**
- **Deployment and production run simulation**
 - Docker image deployment
 - Simulate production run
- **Dataset overview**

ACS Gemini Value Proposition

Simplify the ACS RTDI Experience for Engineers and Customers

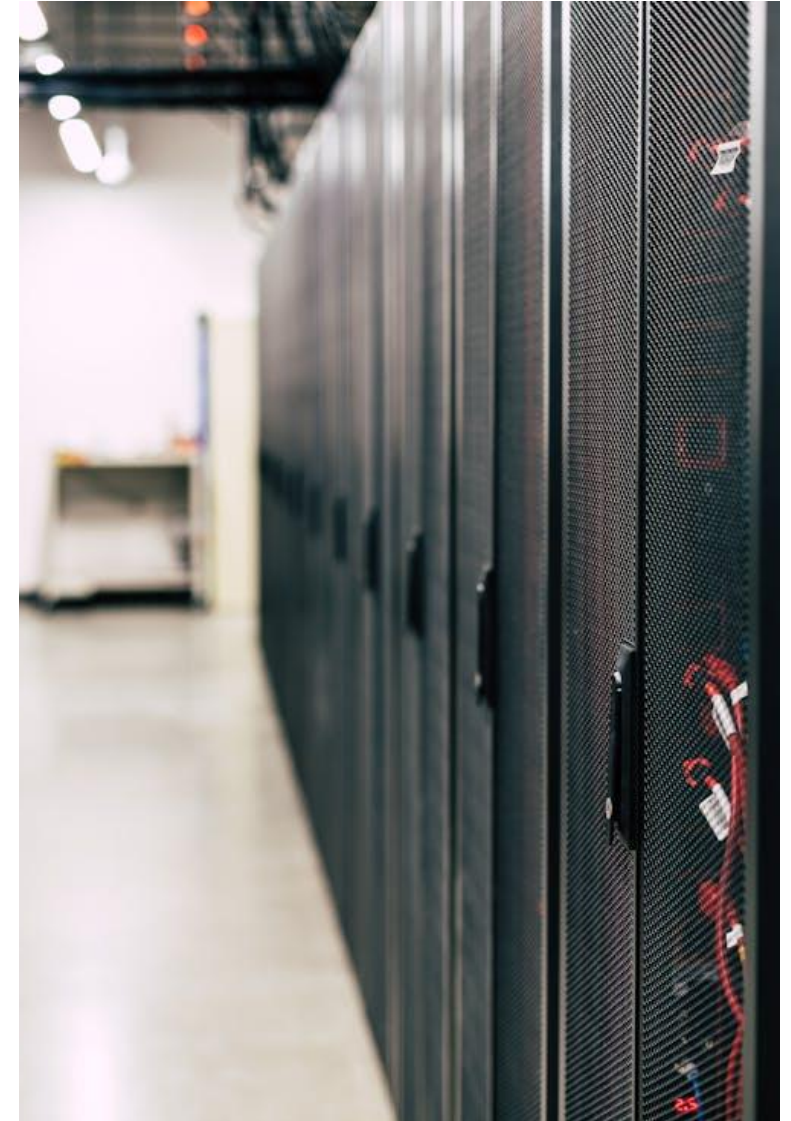
- Accelerate AI/ML algorithm development
- Enable AI/ML app performance benchmarks
- Minimize physical resource investments during the ramp-up phase
- Offer an SDK to integrate with ACS RTDI

Provide a Virtual RTDI Experience for Promotional Purposes

- Showcase ACS RTDI features through product demos
- Provide training for internal users and customer engineers to familiarize them with ACS RTDI.

Expedite customer engagement steps

1. Customer uses ACS Gemini to familiarize with ACS RTDI
2. Customer starts ACS RTDI POC (Proof of concept)
3. Customer implements ACS RTDI in production

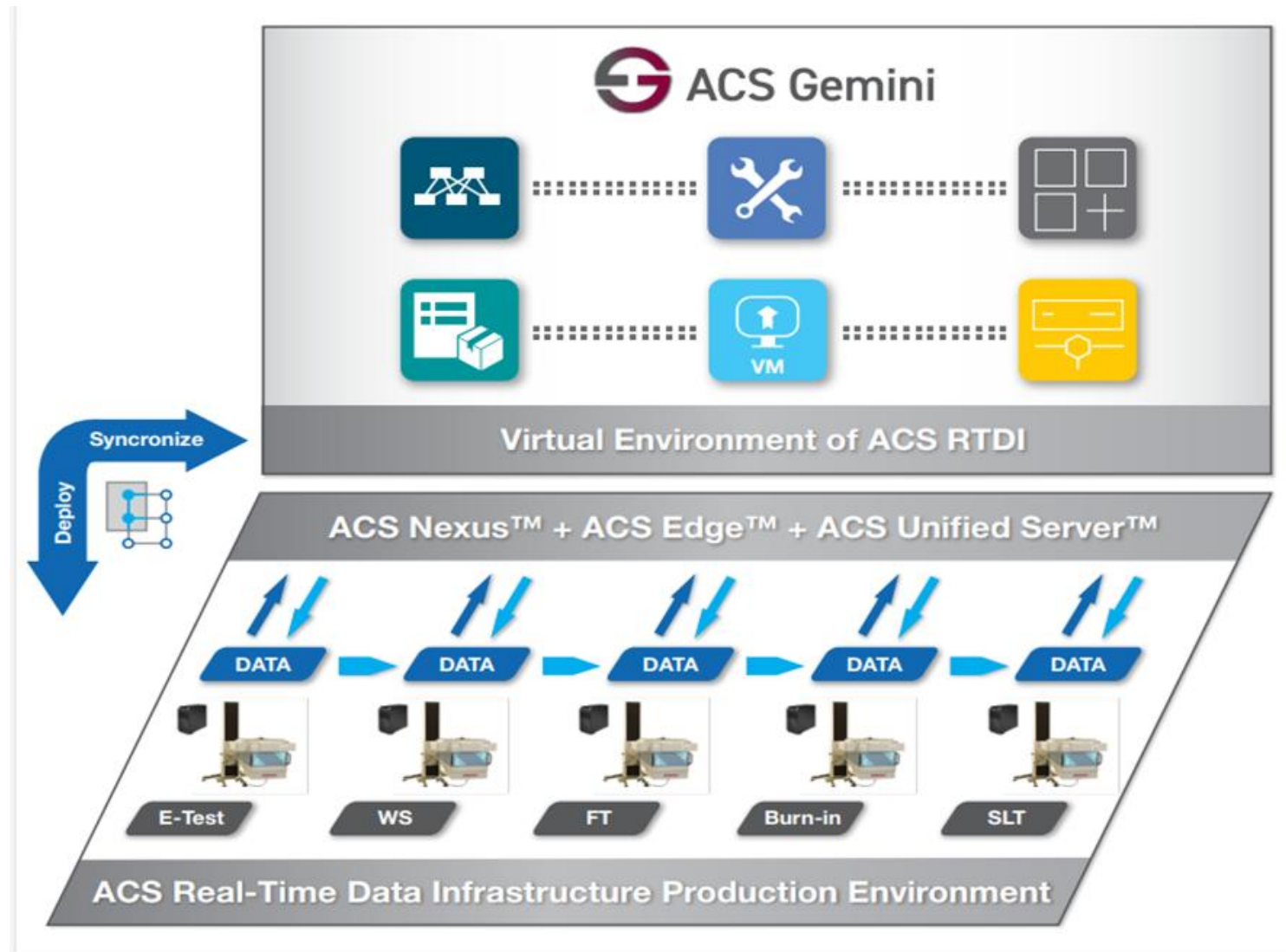


ACS Gemini: How it Works

An environment based on **virtual machines** that replicates a complete ACS RTDI setup, integrating ACS Container Hub, ACS Nexus, ACS Unified Server, and ACS Edge.

Develop apps using synthetic or historical data to evaluate impact on yield and quality across one or more simulated test sessions.

Deploy: After optimization, models can be containerized for streamlined deployment in real production with ACS RTDI.



ACS Gemini User Experience

Remote Desktop (VNC, SSH, SFTP)

- Access the UI/UX through web browser
- Creates virtual machines and a virtualized environment

ACS Applications Reference Model

- Reference model to mimic and expedite the ML development.

ACS Software Development Kit

- Enabling debugging Smart Test (7 or 8) OS Test Programs with synthetic or historical data
- User Manual to create DIY Apps
- STDF replay and feeding for ML modeling and apps performance evaluation

Publish Apps to ACS Container Hub

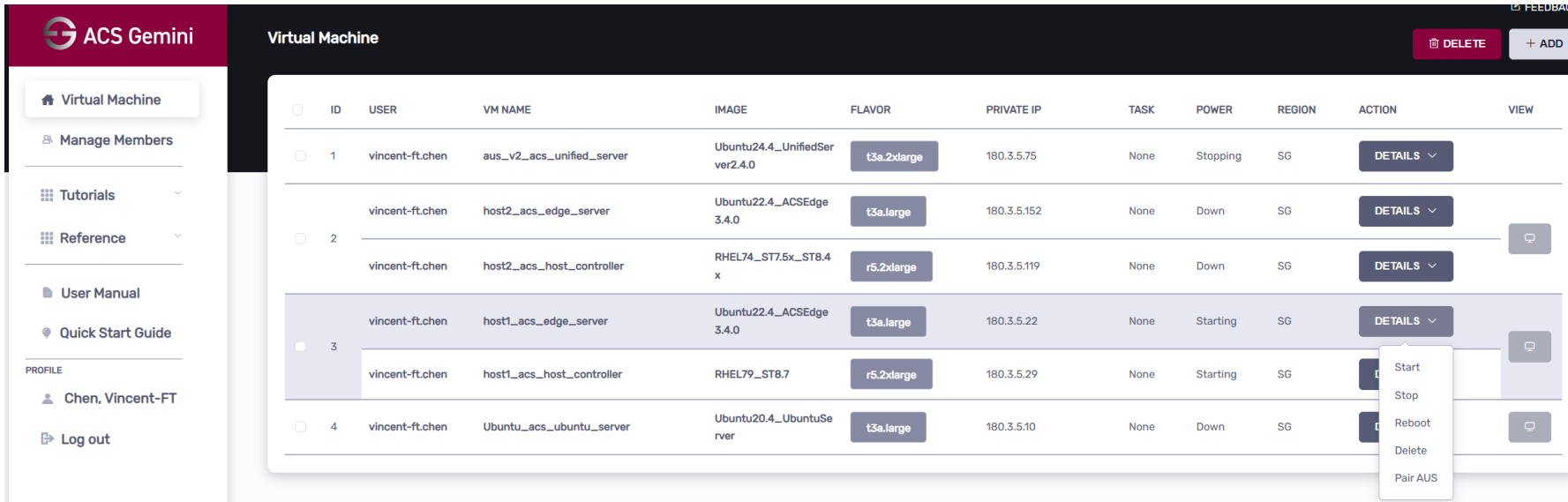
- Zero Trust Security environment
- Applications delivered with high security to a physical ACS RTDI in production after optimization

The screenshot displays the ACS Gemini web interface. On the left is a sidebar menu with options: Virtual Machine, Tutorials, User Manual, Quick Start Guide, and a profile section for 'Chu, Vincent' with a 'Log out' button. The main area is titled 'Virtual Machine List' and contains a table with columns: ID, VM NAME, IMAGE, FLAVOR, PRIVATE IP, TASK, POWER, REGION, ACTION, and VIEW. There are two rows of VMs listed. At the top right of the main area are 'DELETE' and '+ ADD' buttons. At the bottom of the page is a footer with the 'ADVANTEST' logo, links for 'Contact', 'Terms of Use', and 'Privacy Policy', and copyright information: '@ ADVANTEST CORPORATION. All Rights Reserved ACS Gemini V1.2.1.0'.

ID	VM NAME	IMAGE	FLAVOR	PRIVATE IP	TASK	POWER	REGION	ACTION	VIEW
1	demoaus_acs_unified_server	Ubuntu22.4_UnifiedServer1.2.0-prod	t3a.large	172.173.125.215	None	Up	US	DETAILS	
2	demohc_acs_edge_server	Ubuntu22.4_ACSEdge3.0.3-prod	t3a.large	172.173.125.86	None	Up	US	DETAILS	
	demohc_acs_host_controller	RHEL74_ST7.5x_ST8.4x	r5xlarge	172.173.125.179	None	Up	US	DETAILS	

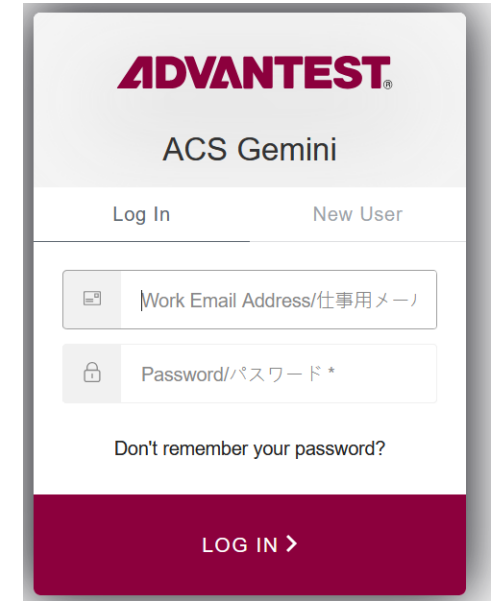
How to connect to Gemini

- **Use browser (firefox/chrome/edge) to connect to ACS Gemini**
 - URL: [ACS Gemini](https://sandbox.gemini.te-cloud.advantest.com/dashboard/virtual-machine)(https://sandbox.gemini.te-cloud.advantest.com/dashboard/virtual-machine)
 - The account and passwd is needed.
 - Account: eventgroup0919@gmail.com
 - Pass: 1Qaz2wsx
- **After login, user can find all VMs for this account.**



The screenshot shows the ACS Gemini Virtual Machine dashboard. On the left is a sidebar with navigation links: Virtual Machine, Manage Members, Tutorials, Reference, User Manual, and Quick Start Guide. Below these is a profile section for 'Chen, Vincent-FT' with a 'Log out' button. The main area is titled 'Virtual Machine' and contains a table of VMs. The table has columns for ID, USER, VM NAME, IMAGE, FLAVOR, PRIVATE IP, TASK, POWER, REGION, ACTION, and VIEW. There are four VMs listed. A context menu is open over the third VM, showing options: Start, Stop, Reboot, Delete, and Pair AUS.

ID	USER	VM NAME	IMAGE	FLAVOR	PRIVATE IP	TASK	POWER	REGION	ACTION	VIEW
1	vincent-ft.chen	aus_v2_acs_unified_server	Ubuntu24.4_UnifiedServer2.4.0	t3a.2xlarge	180.3.5.75	None	Stopping	SG	DETAILS	
2	vincent-ft.chen	host2_acs_edge_server	Ubuntu22.4_ACSEdge3.4.0	t3a.large	180.3.5.152	None	Down	SG	DETAILS	
	vincent-ft.chen	host2_acs_host_controller	RHEL74_ST7.5x-ST8.4x	r5.2xlarge	180.3.5.119	None	Down	SG	DETAILS	
3	vincent-ft.chen	host1_acs_edge_server	Ubuntu22.4_ACSEdge3.4.0	t3a.large	180.3.5.22	None	Starting	SG	DETAILS	
	vincent-ft.chen	host1_acs_host_controller	RHEL79_ST8.7	r5.2xlarge	180.3.5.29	None	Starting	SG		
4	vincent-ft.chen	Ubuntu_acs_ubuntu_server	Ubuntu20.4_UbuntuServer	t3a.large	180.3.5.10	None	Down	SG		



The login form for ACS Gemini. It features the ADVANTEST logo and 'ACS Gemini' text. There are links for 'Log In' and 'New User'. The form includes input fields for 'Work Email Address/仕事用メール' and 'Password/パスワード *'. A 'Don't remember your password?' link is present. A large red 'LOG IN >' button is at the bottom.

How to access VM

Step 1. Press the button of VM which want to access

Virtual Machine

DELETE

+ ADD

☐	ID	USER	VM NAME	IMAGE	FLAVOR	PRIVATE IP	TASK	POWER	REGION	ACTION	VIEW
☐	1	vincent-ft.chen	aus_v2_acs_unified_server	Ubuntu24.4_UnifiedServer24.0	t3a.2xlarge	180.3.5.75	None	Down	SG	DETAILS	
☐	2	vincent-ft.chen	host2_acs_edge_server	Ubuntu22.4_ACSEdge3.4.0	t3a.large	180.3.5.152	None	Down	SG	DETAILS	
		vincent-ft.chen	host2_acs_host_controller	RHEL74_ST7.5x_ST8.4x	r5.2xlarge	180.3.5.119	None	Down	SG	DETAILS	
☐	3	vincent-ft.chen	host1_acs_edge_server	Ubuntu22.4_ACSEdge3.4.0	t3a.large	180.3.5.22	None	Up	SG	DETAILS	
		vincent-ft.chen	host1_acs_host_controller	RHEL79_ST8.7	r5.2xlarge	180.3.5.29	None	Up	SG	DETAILS	
☐	4	vincent-ft.chen	Ubuntu_acs_ubuntu_server	Ubuntu20.4_UbuntuServer	t3a.large	180.3.5.10	None	Down	SG	DETAILS	

Step 2. Select protocol to access VM

 ACS Gemini

RECENT CONNECTIONS



VNC



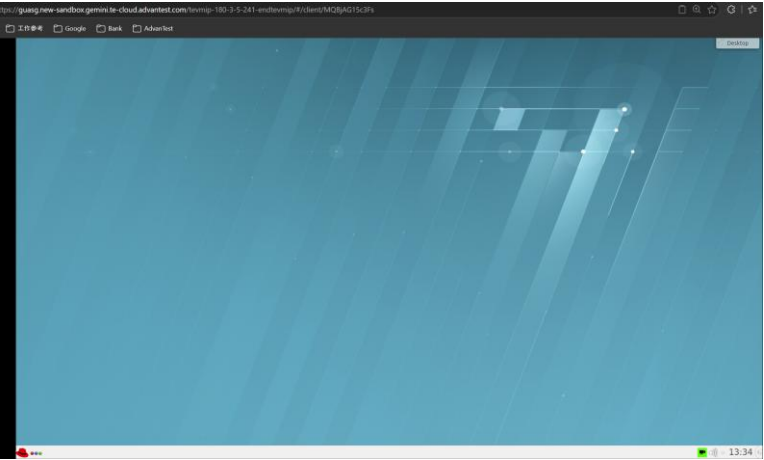
SSH

ALL CONNECTIONS

SSH

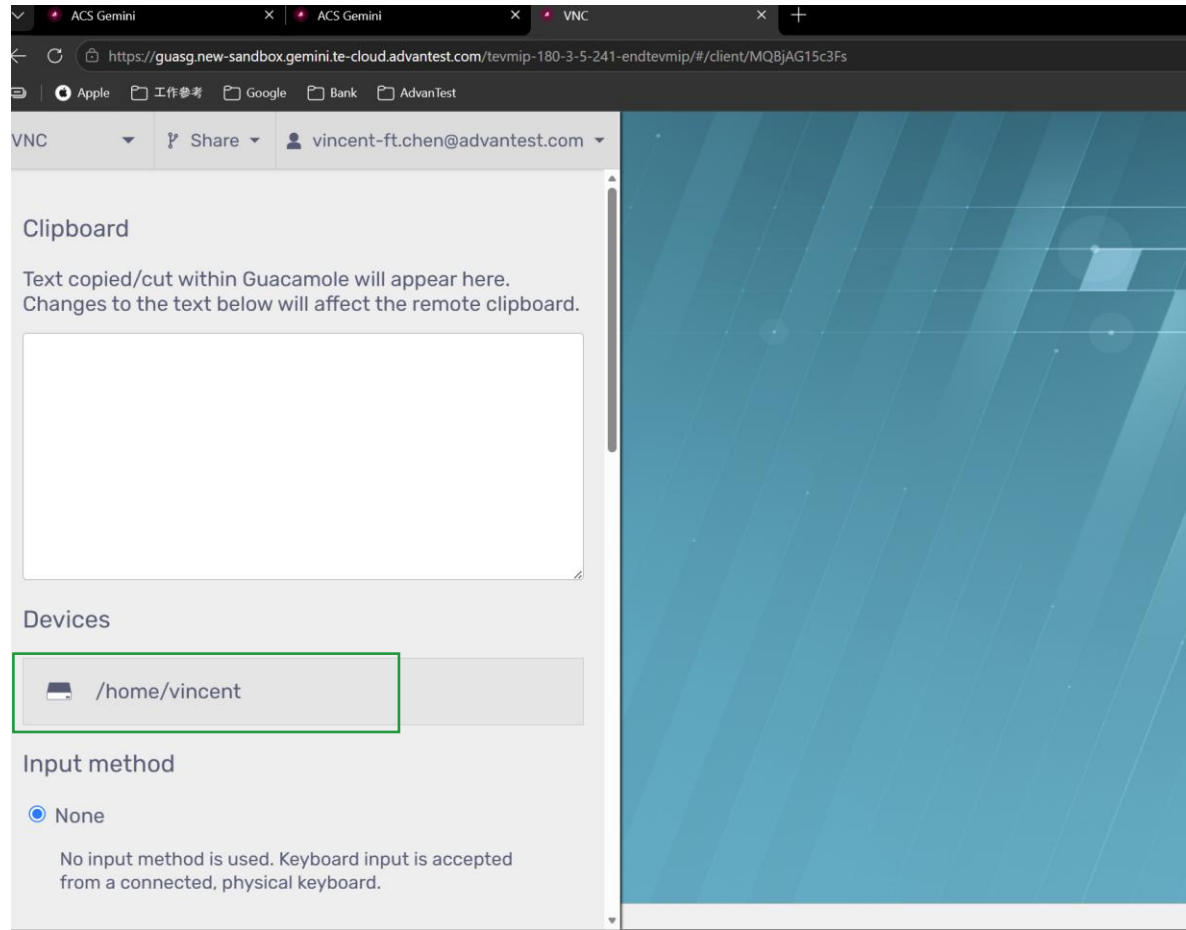
VNC

Step 3. Use RHEL 7.9 VM

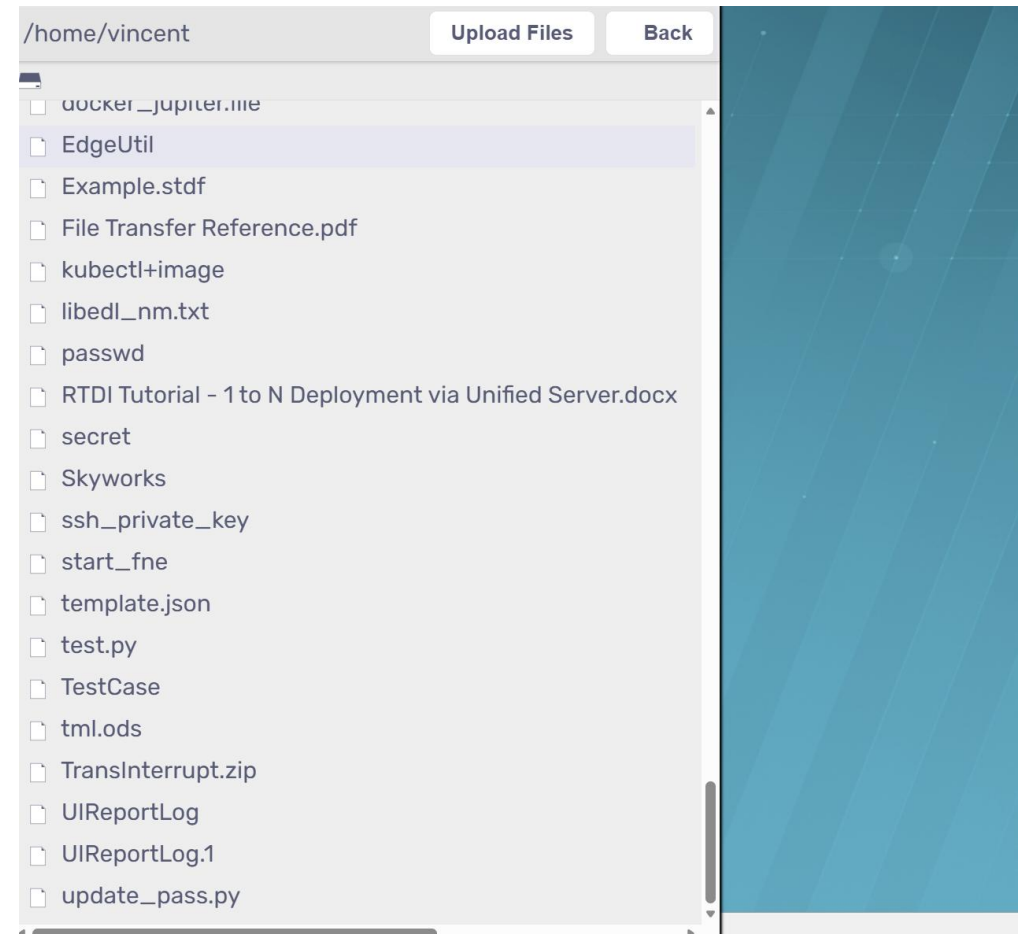


File upload/download to VM

Press Shift+Ctrl+Alt in the VM
Click the devices



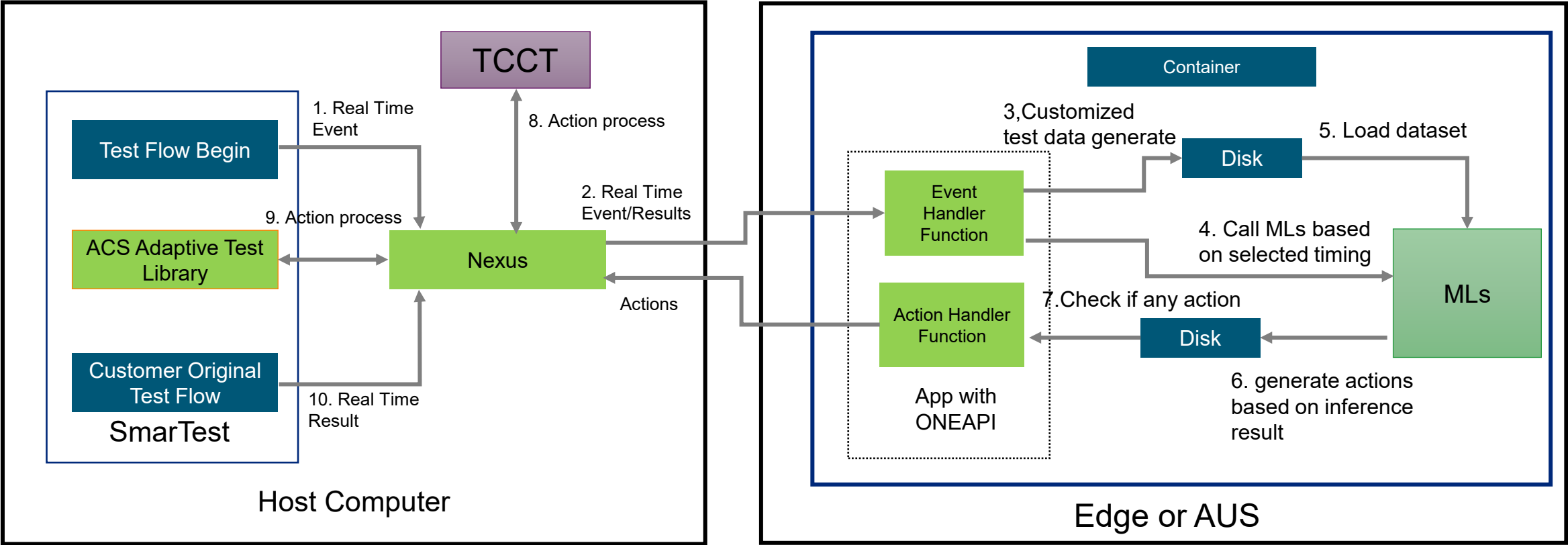
Press Upload Files to upload files
Or double click the target file to download to PC.



Agenda

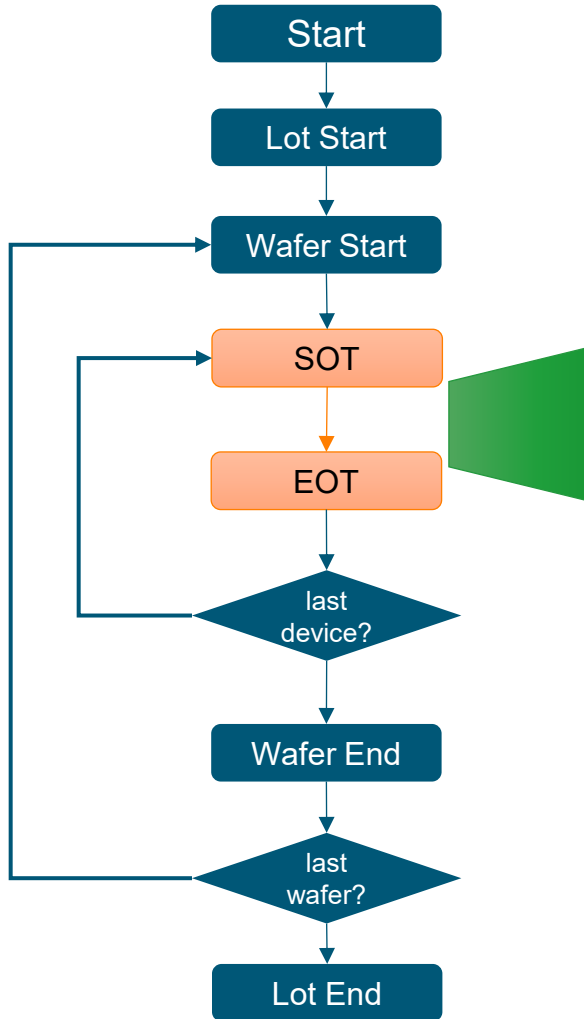
- **What is RTDI**
- **Gemini overview**
 - How to use Gemini
- **NEXUS ONEAPI overview**
- **Development environment**
- **Deployment and production run simulation**
 - Docker image deployment
 - Simulate production run
- **Dataset overview**

Nexus/ONEAPI overview

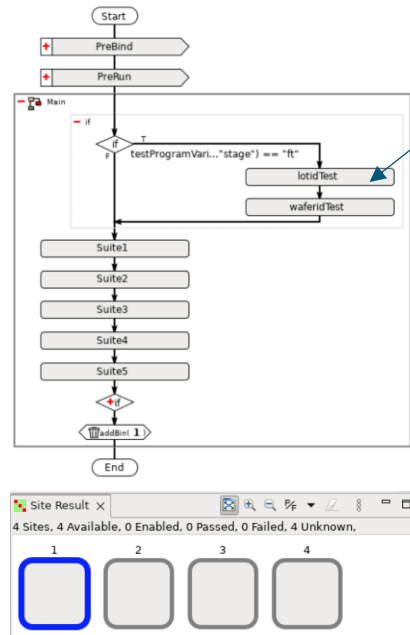


Step 3-6 will be different based on requirement.

Production event and test results



test program

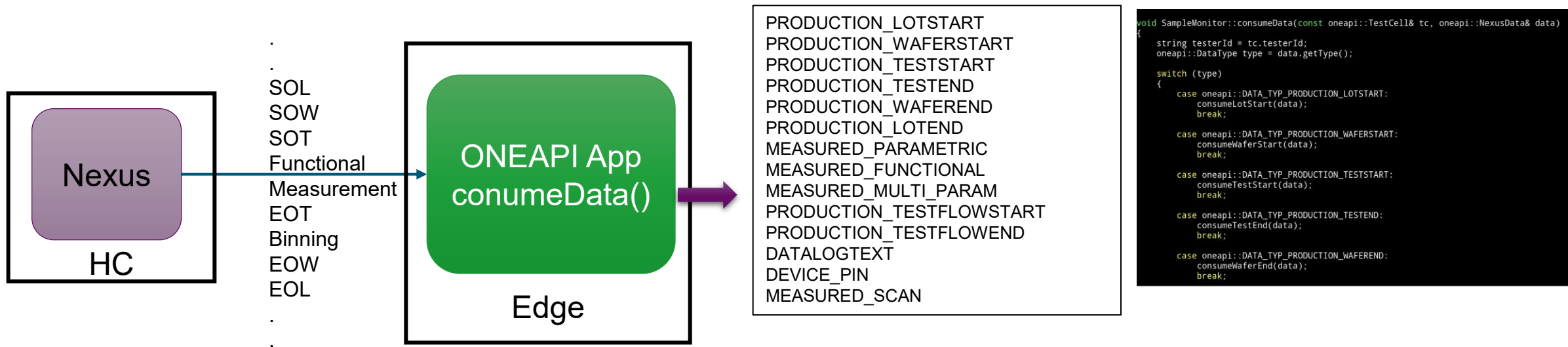


Site to run test flow

- Each block means one test suite.
 - Each test suite may contain several test results.
 - Type of test result
 - Functional test – pass /fail only
 - Parametric test – one measure value, float or double
 - Multiple parametric test – results of multiple pins, float or double.
 - Scan test –pass/fail only
 - Each record of test result will contain test suite, test name and test number, and multiple parametric test has additional pin information.
- After test program finish execution, each device be assigned the soft bin number and hard bin number based on test results of each site.

ACS ONEAPI Overview

- ONEAPI provide a standard interface to receive real time data with standard format during production.
 - ✓ Nexus will communicate with SmarTest to receive real time data and forward the data to app which is using ONEAPI.
- Each test result event will trigger function consumeData in app which is developed by ONEAPI.
 - ✓ The app can be processing/analyze data after receive the test results from Nexus.



Overview of code

- Interface.registerMonitor register the class which contain consumeData
- Interface.connect start to monitor test event from Nexus.
 - When there is any event arrive, the ONEAPI will call consumeData and occupied the thread until the function call finished.

```
main.py > main
33
34 def main():
35     configure logging()
36     myMonitor = SampleMonitor()
37     Interface.registerMonitor(myMonitor)
38
39     me = AppInfo()
40     me.name = "sample" # Recommend using hostname (hostname = socket.gethostname())
41     me.vendor = "adv"
42     me.version = "1.0.0"
43
44     NexusDataEnabled = True # whether to enable Nexus Data Streaming and Control
45     TPServiceEnabled = True # whether to enable TPService for communication with NexusTP
46     res = Interface.connect(me, NexusDataEnabled, TPServiceEnabled)
47
48     if res != 0:
49         logging.error(f"Connection request failed to initiate. code = {res}")
50         sys.exit()
51
52     logging.info("Connection request initiated successfully.")
53
54     signal.signal(signal.SIGINT, quit)
55     signal.signal(signal.SIGTERM, quit)
56
57     while True:
58         signal.pause()
```

```
main.py • sample.py x
sample.py > ...
463
464 def consumeData(self, tc, data):
465     print(f"===== consume data from: testerId = {tc.testeId} =====")
466     datatype = data.getType()
467     if datatype == DataType.DATA_TYP_PRODUCTION_LOTSTART:
468         self.consumeLotStart(data)
469     elif datatype == DataType.DATA_TYP_PRODUCTION_LOTEND:
470         self.consumeLotEnd(data)
471     elif datatype == DataType.DATA_TYP_PRODUCTION_WAFERSTART:
472         self.consumeWaferStart(data)
473     elif datatype == DataType.DATA_TYP_PRODUCTION_WAFEREND:
474         self.consumeWaferEnd(data)
475     elif datatype == DataType.DATA_TYP_PRODUCTION_TESTSTART:
476         self.consumeTestStart(data)
477     elif datatype == DataType.DATA_TYP_PRODUCTION_TESTEND:
478         self.consumeTestEnd(data)
479     elif datatype == DataType.DATA_TYP_PRODUCTION_TESTFLOWSTART:
480         self.consumeTestFlowStart(data)
481     elif datatype == DataType.DATA_TYP_PRODUCTION_TESTFLOWEND:
482         self.consumeTestFlowEnd(data)
483     elif datatype == DataType.DATA_TYP_MEASURED_PARAMETRIC:
484         self.consumeParametricTest(data)
485     elif datatype == DataType.DATA_TYP_MEASURED_FUNCTIONAL:
486         self.consumeFunctionalTest(data)
487     elif datatype == DataType.DATA_TYP_MEASURED_MULTI_PARAM:
488         self.consumeMultiParametric(data)
489     elif datatype == DataType.DATA_TYP_DEVICE:
490         self.consumeDeviceData(data)
491     elif datatype == DataType.DATA_TYP_USERDEFINED:
492         self.consumeUserDefinedData(data)
493     elif datatype == DataType.DATA_TYP_DATALOGTEXT:
```

ONEAPI User Guide and example log

ADVANTEST

ADVANTEST CORPORATION

ACS Real-Time
Data Infrastructure (RTDI)
ONEAPI User Guide

This version includes:
ACS Nexus v3.2.0



© 2023-2026 ADVANTEST Corporation
All rights reserved

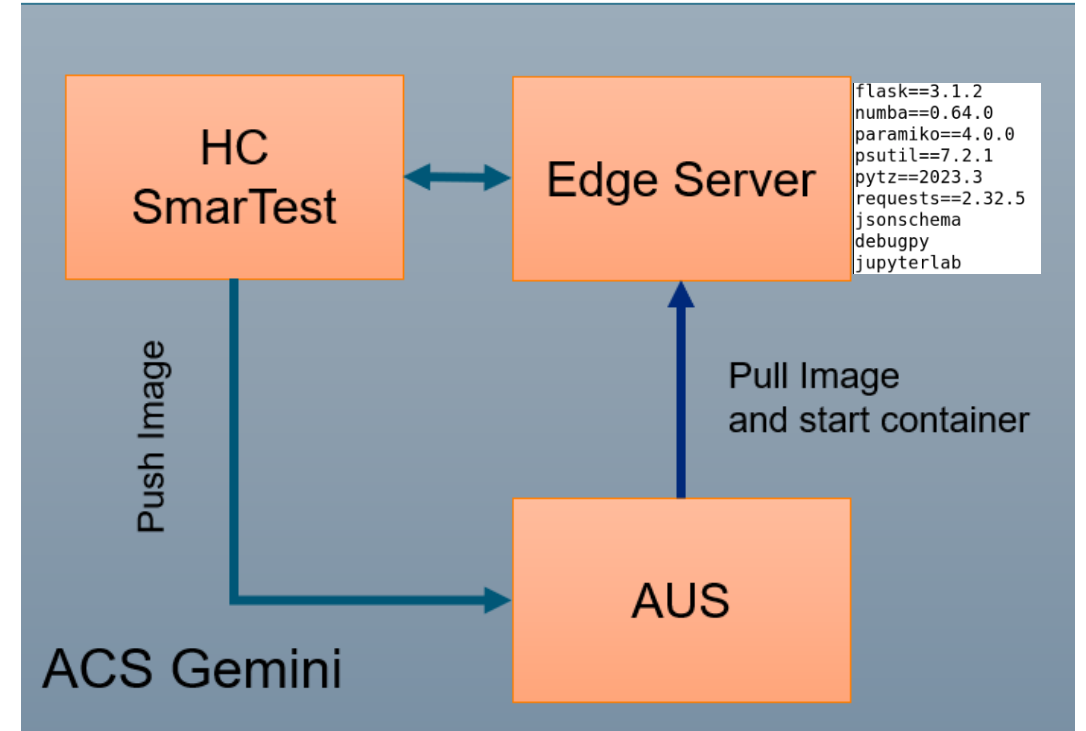
[Abstract]
Printed in U.S.A

Agenda

- **What is RTDI**
- **Gemini overview**
 - How to use Gemini
- **NEXUS ONEAPI overview**
- **Development environment**
- **Deployment and production run simulation**
 - Docker image deployment
 - Simulate production run
- **Dataset overview**

Development environment

- **Edge server:**
 - When develop solution which need to use ONEAPI, user can use firefox in host controller to connect vscode server with password to development environment in Edge server.
 - The vs code server provide python 3.10.
 - **Development environment is created with a docker and there is persist volume which folder is /home/debugger/project to avoid file missing once the container stop or crash.**
- **Host Controller:**
 - SmarTest in HC is in charged to communicate with tester, perform the measurement and generate test results to data log stream.
- **AUS:**
 - After solution finishes development, user should build docker images in host controller and push to AUS. During production run simulation, the Edge server will pull the image from AUS and start the image to wait test result event from host controller.



Engineer package detail

- There are several script of application to support user to reduce the effort to use SmarTest or deployed application log check.

- Edge:

- EdgeLog is the directory which save a utility to query partial log from edge server.

```
[vincent@host1 Edge]$ ./EdgeLog/EdgeLog Log
2026-08-14 10:58:39 Info EdgeLog V1.0.0 2026/08/13
2026-08-14 10:58:39 Info Host ip: 180.3.5.29
2026-08-14 10:58:39 Info Edge Verion: ACS Edge Server API-3.4.0
2026-08-14 10:58:40 Info ***** Fetech Containers Log *****
container: py-app
[2026-08-14 02:07:07,382] [INFO] [configuration.cpp:135] ===== ACS ONEAPI 3.3.0.rc2 =====
[2026-08-14 02:07:07,382] [INFO] [configuration.cpp:35] load configuration...
[2026-08-14 02:07:07,382] [INFO] [configuration.cpp:36] [Configuration] ONEAPI DEBUG : 6
```

- oneAPI_py3.10 is example project which utilize ONEAPI to collect the real time data.

- SmarTest

- Case_Smt870 is SmarTest program to simulate production.
 - recipe save all necessary files to simulate production.
 - runTp.sh is the script to help to load/run test program/run production.
 - Util keep all required script for runTp.sh
 - app_descriptor.josn:
 - Define what image should be use, the container name after start...etc.

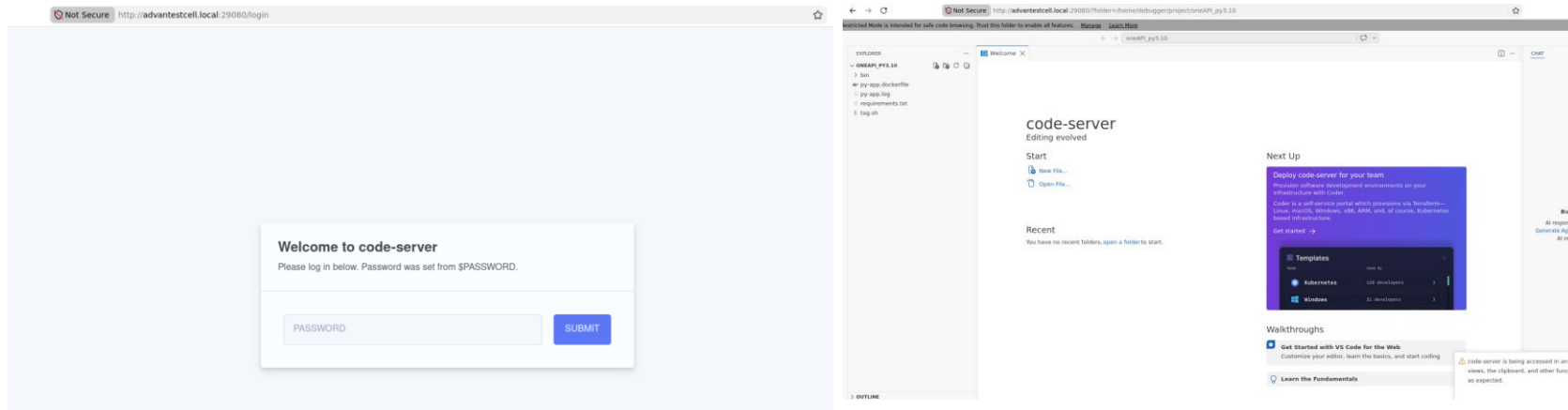
- firefox_128

- The newer browser to support development.

```
Case_Event/
|-- Edge
|   |-- EdgeLog
|   |-- oneAPI_py3.10
|-- SmarTest
|   |-- Case_Smt870
|   |-- Util
|   |-- app_descriptor.json
|   |-- recipe
|   |-- runTp.sh
|   |-- workspace
|-- firefox_128
|   |-- active-update.xml
|   |-- application.ini
|   |-- browser
|   |-- crashhelper
|   |-- crashreporter
```

How to access development environment

- From host control, user can use ssh or firefox to connect development environment and passwd is needed.
- Url: <http://advantestcell.local:29080> passwd: 1qaz2wsx



- ssh: `ssh -p 29022 debugger@advantestcell.local`

```
[vincent@host1 ~]$ ssh -p 29022 debugger@advantestcell.local
debugger@advantestcell.local's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1061-aws x86_64)
```

```
* Documentation: https://help.ubuntu.com
* Management:   https://landscape.canonical.com
* Support:       https://ubuntu.com/pro
```

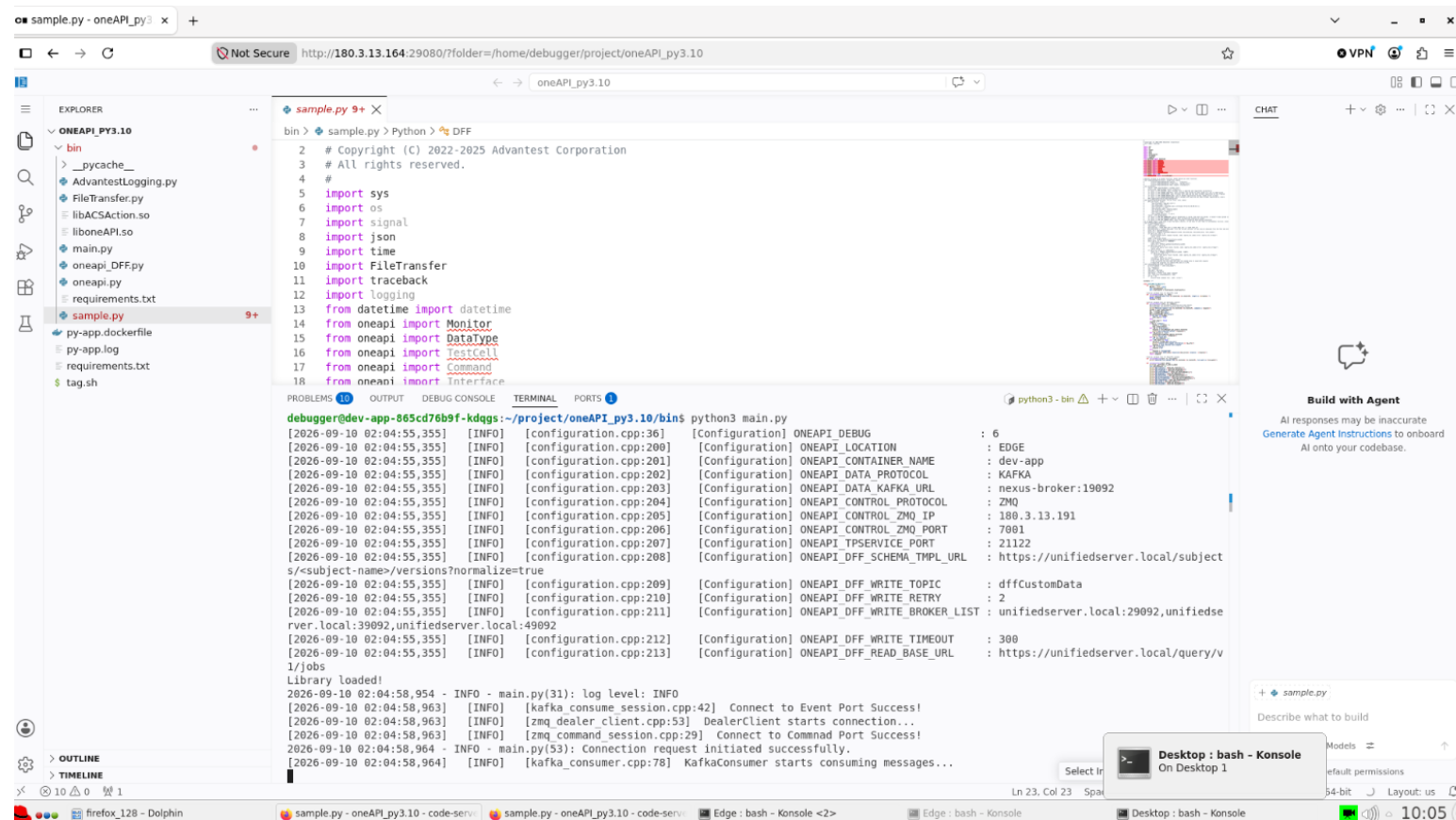
This system has been minimized by removing packages and content that are not required on a system that users do not log into.

```
To restore this content, you can run the 'unminimize' command.
Last login: Fri Aug 14 03:19:30 2026 from 10.42.0.1
debugger@debug-app-6fd6fc78ff-2ctnq:~$ ls
project  update_pass.py  workspace
debugger@debug-app-6fd6fc78ff-2ctnq:~$ ll
total 72
drwxr-xr-x 1 debugger debug_grp 4096 Aug 14 03:23 ./
drwxr-xr-x 1 root      root      4096 Aug 12 07:58 ../
-rw-r--r-- 1 debugger debug_grp   5 Aug 14 03:19 .bash_history
-rw-r--r-- 1 debugger debug_grp  220 Jan  6 2022 .bash_logout
-rw-r--r-- 1 debugger debug_grp 11199 Aug 14 00:40 .bashrc
drwxr-xr-x 4 debugger debug_grp 4096 Aug 14 03:19 .cache/
```

- User can use sftp or vs code server to upload/download file.

How to run the code in development environment

- After uploading the oneAPI template package to development environment, use python3 to run main.py



```
oneAPI_py3.10/  
-- bin  
| -- AdvantestLogging.py  
| -- FileTransfer.py  
| -- __pycache__  
| | -- oneapi.cpython-310.pyc  
| | -- oneapi_DFF.cpython-310.pyc  
| -- libACSAAction.so  
| -- liboneAPI.so  
| -- main.py  
| -- oneapi.py  
| -- oneapi_DFF.py  
| -- requirements.txt  
| -- sample.py  
-- py-app.dockerfile  
-- py-app.log  
-- requirements.txt  
-- tag.sh
```

How to trigger SmarTest load program and run manually

- When user want to debug application which is using ONEAPI to receive real time result from SmarTest, user need to run the program manually.
- runTp.sh can help user to load test program/ run program manually or run production simulation.
 - Usage: ./runTp.sh {load|eng_run|prod_run} [loop_count]
 - “./runTp.sh load” will start SmarTest and load the target test program.
 - “./runTp.sh eng_run 2” will run test program two times.
 - “./runTp.sh prod_run” will run production simulation.

```
[vincent@host1 SmarTest]$ ./runTp.sh load
command =/home/vincent/Case_Event/SmarTest/Util/startSmt.py
2026-08-14 05:13:54,729 - INFO - workspace: /home/vincent/Case_Event/SmarTest/workspace
run mode: eng
2026-08-14 05:13:55,058 - INFO - run cmd: ['/opt/hp93000/soc/prod_env/sbin/kill_smarTest'] Success
2026-08-14 05:14:24,591 - INFO -
2026-08-14 05:14:24,592 - INFO - Found 0 running PH sessions.
2026-08-14 05:14:24,592 - INFO - Checking remaining PH session processes.
2026-08-14 05:14:24,592 - INFO - No remaining processes. Done!
2026-08-14 05:14:24,592 - INFO - Errors: b'/opt/hp93000/testcell/sbin/kill_tc_sessions.sh: line 119: kill: (7656) - Operation
not permitted\nSmarTest no longer running for current user. Cleaning up.\n'
2026-08-14 05:14:24,592 - INFO - run cmd: ['/opt/hp93000/testcell/bin/killTcct.prod'] Success
2026-08-14 05:14:55,856 - INFO - run cmd: /opt/hp93000/soc/prod_env/bin/HPSmarTest --data=/home/vincent/Case_Event/SmarTest/wo
rkspace --testprogram=TestCase_Smt870/src/TestCase1/TestCase1_4site_ft.prog -o& Success
2026-08-14 05:15:25,939 - INFO - [INFO] Starting 1:1 app deployment...
2026-08-14 05:15:25,939 - INFO - Errors: b'[ERROR] Request failed.\n[ERROR] Session is not ready, cannot start or stop applica
tion.\n'
2026-08-14 05:15:25,940 - INFO - run cmd: ['/opt/acs/nexus/bin/AppDeployer start'] Fail
2026-08-14 05:16:26,533 - INFO - [INFO] Starting 1:1 app deployment...
2026-08-14 05:16:26,533 - INFO - [INFO] Request completed successfully.
2026-08-14 05:16:26,533 - INFO - run cmd: ['/opt/acs/nexus/bin/AppDeployer start'] Success
You have mail in /var/spool/mail/vincent
[vincent@host1 SmarTest]$ ./runTp.sh eng_run 1
command =/home/vincent/Case_Event/SmarTest/Util/tpExec
test program: TestCase_Smt870/src/TestCase1/TestCase1_4site_ft.prog validate mode: Run 1/1
Site : 1 , sbin: 1 , sbin name: passed , hbin: 1
Site : 2 , sbin: 1 , sbin name: passed , hbin: 1
Site : 3 , sbin: 1 , sbin name: passed , hbin: 1
Site : 4 , sbin: 1 , sbin name: passed , hbin: 1
```

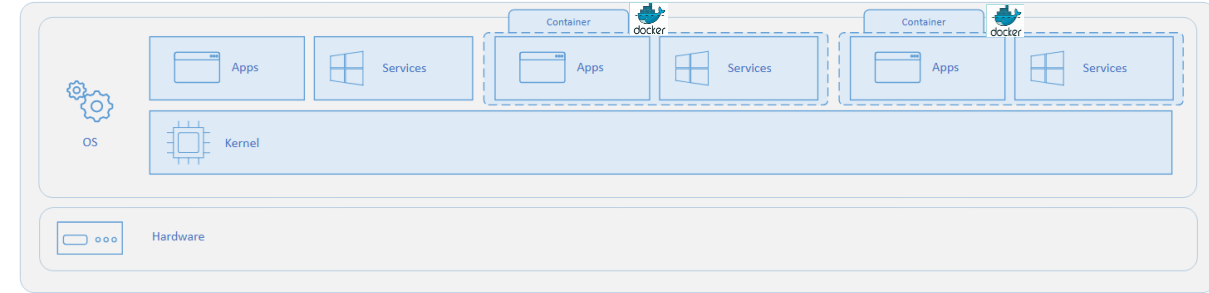
Two steps to trigger program run.

1. If program is not loaded, run ./runTp.sh load first.
2. If program is loaded, run ./runTp.sh eng_run 1

Docker file example for image build

■ What is docker images

- A Docker image is a read-only template that contains everything needed to run an application:
- Application code
- Runtime (Java, Python, Node.js, etc.)
- Libraries and dependencies
- Environment variables
- Configuration files
- Think of it like a snapshot or blueprint of an application.



■ Docker file is defined how to build the image to support user's requirement.

- Example:

```
FROM unifiedserver.local/acs-app/template-data-app:v22.04
```

Base image

```
WORKDIR /opt/nexus/OneAPI
```

Change directory in image

```
COPY bin/. ./bin
```

Copy file to image

```
ENV ONEAPI_DEBUG 6
```

Set environment variable in image

```
WORKDIR /opt/nexus/OneAPI/bin
```

```
CMD ["python3", "-u", "main.py"]
```

Run the script in images

How to deploy image to AUS

- **tag.sh** is a script that will do image build based on py-app.dockerfile, push image to AUS.
- **Need to run tag.sh with root permission.**
- `sudo ./tag.sh`

```
[vincent@host1 oneAPI_py3.10]$ sudo ./tag.sh
Error: No such image: unifiedserver.local/acs-app/py-app:latest
[+] Building 0.4s (9/9) FINISHED
=> [internal] load build definition from py-app.dockerfile
=> => transferring dockerfile: 236B
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load metadata for unifiedserver.local/acs-app/tem
=> [1/4] FROM unifiedserver.local/acs-app/template-data-app:v22
=> [internal] load build context
=> => transferring context: 12.62MB
=> CACHED [2/4] WORKDIR /opt/nexus/OneAPI
```

```
oneAPI_py3.10/
|-- bin
|   |-- AdvantestLogging.py
|   |-- FileTransfer.py
|   |-- __pycache__
|       |-- oneapi.cpython-310.pyc
|       |-- oneapi_DFF.cpython-310.pyc
|   |-- libACSAction.so
|   |-- liboneAPI.so
|   |-- main.py
|   |-- oneapi.py
|   |-- oneapi_DFF.py
|   |-- requirements.txt
|   |-- sample.py
|-- py-app.dockerfile
|-- py-app.log
|-- requirements.txt
|-- tag.sh
```



```
#!/bin/bash
#set -x
orginame="py-app"
iname="py-app:latest"
registry="unifiedserver.local/acs-app"
docker rmi $registry/$iname
docker build -f py-app.dockerfile . -t $registry/$iname
docker push $registry/$iname
docker rmi $registry/$iname
```

Simulate production run

- Execute “./runTp.sh prod_run” to run production simulation.

The screenshot displays the V93000 Test Cell Control Tool-8.7.5.2 interface. The top status bar shows Tester Status: TESTPROGRAM_BOUND, Equipment Status: (empty), and Recipe Status: IDLE. The main workspace contains a grid of Site1, Site2, Site3, and Site4, each with a 2x2 matrix of values (1, 1, 1, 1) and a bottom row of values (77, 78, 79, 80). The ACS NEXUS 3.3.0 status window shows Session: SMT8 Ready, Host Controller: Disabled, ACS Edge: Available, and ACS Unified Server: Available. A terminal window titled 'SmarTest : runTp.sh - Konsole' shows the execution of the command './runTp.sh prod_run' and the resulting output, including workspace information and kill commands for SmarTest and Tcct.

Warning configure

```
SmarTest : runTp.sh - Konsole
File Edit View Bookmarks Settings Help
[vincent@host1 Edge]$ cd ..
[vincent@host1 Case_Event]$ cd SmarTest/
[vincent@host1 SmarTest]$ ls
Case Smt870 Util app_descriptor.json recipe runTp.sh workspace
[vincent@host1 SmarTest]$ ./runTp.sh prod_run
command =/home/vincent/Case_Event/SmarTest/Util/startSmt.py prod
2026-09-10 02:16:35,334 - INFO - workspace: /home/vincent/Case_Event/SmarTest/
run mode: prod
2026-09-10 02:16:35,334 - INFO - Kill SmarTest
2026-09-10 02:16:35,892 - INFO - run cmd: ['/opt/hp93000/soc/prod_env/sbin/kill
2026-09-10 02:16:35,892 - INFO - Kill Tcct
```

bin : bash SmarTest : runTp.sh

Agenda

- What is RTDI
- Gemini overview
 - How to use Gemini
- NEXUS ONEAPI overview
- Development environment
- Deployment and production run simulation
 - Docker image deployment
 - Simulate production run
- **Dataset overview**

Dataset

Device Info

Device test results

PID	Lot	Wafer	Site	X	Y	PF	SBin	HBin	Test Time	220_Main.Suite1#CP	240_Main.Suite1_0#DS0	260_Main.Suite2#IO1	280_Main.Suite3#IO2	300_Main.Suite4#IO3	320_Main.Suite5#IO4	340_Main.Suite6#IO5	360_Main.Suite7#IO6	380_Main.Suite8#IO7
Pin										CP	DS0	IO1	IO2	IO3	IO4	IO5	IO6	IO7
Test Num										220	240	260	280	300	320	340	360	380
High Limit										0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Low Limit										1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8
1	A12345	1	1	8	7	0	1	1	42028	1.251	1.17	1.148	1.188	1.218	1.209	1.433	1.383	1.024
2	A12345	1	2	7	6	0	1	1	41369	1.28	1.17	1.22	1.265	1.302	1.124	1.445	1.396	1.1
3	A12345	1	3	1	3	0	1	1	42583	1.297	1.152	1.229	1.217	1.282	1.195	1.44	1.405	1.15
4	A12345	1	4	3	1	0	1	1	36044	1.243	1.13	1.198	1.155	1.267	1.122	1.416	1.366	1.103
5	A12345	1	1	2	5	0	1	1	42954	1.337	1.123	1.162	1.13	1.274	1.088	1.371	1.378	1.067
6	A12345	1	2	10	6	0	1	1	36573	1.304	1.192	1.204	1.156	1.239	1.137	1.468	1.356	1.103

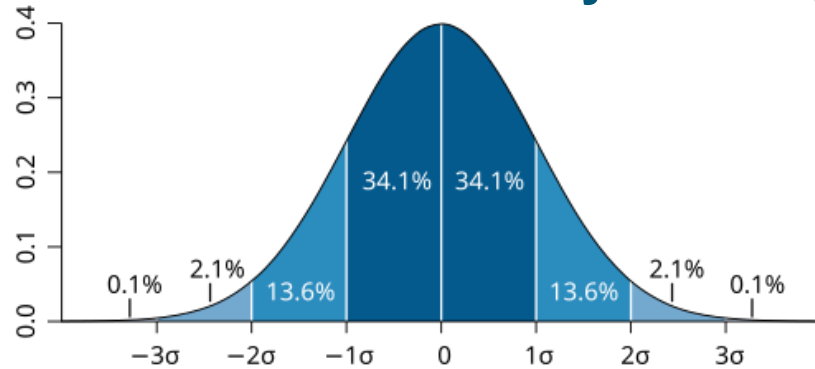
The naming rule of column name of device test results is <test number>_<test suite name>#<pin name>



example.csv

Anomaly Behavior

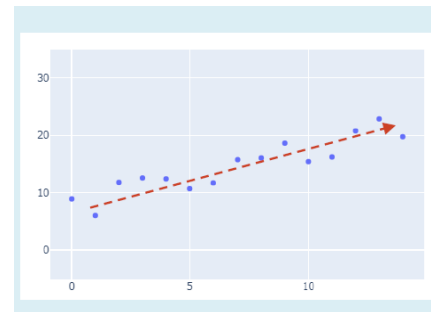
- In production, the distribution of test result will be Gaussian distribution when there is no any anomaly in test cell.



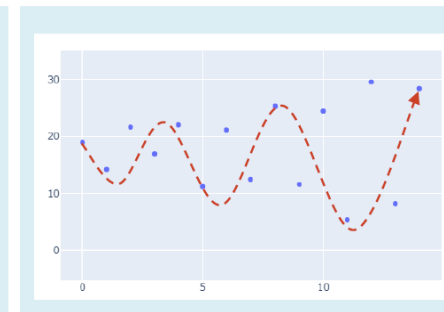
- Site 2 Site unbalance



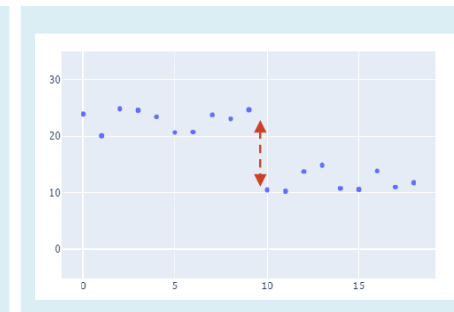
- Trend Issue



Trend up or down



Standard deviation trend change



Measure value shifts

Q&A